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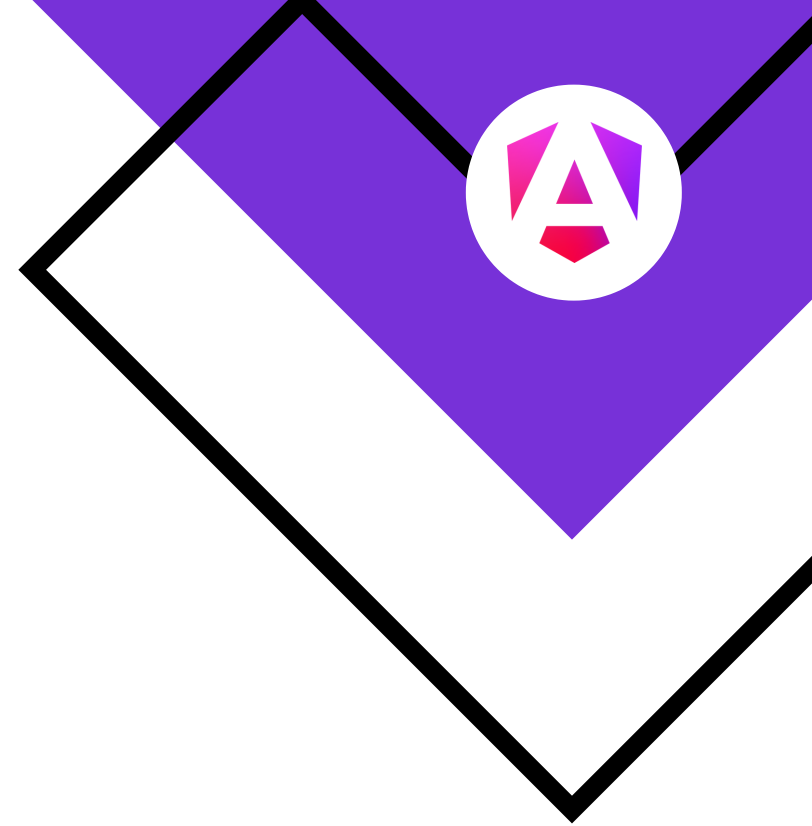
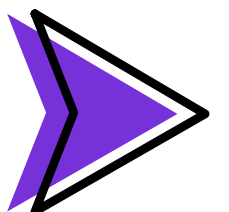
Understanding Angular Architecture:

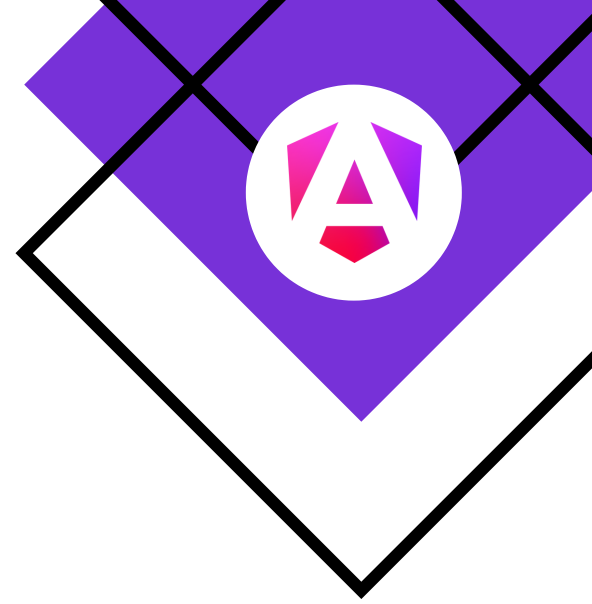
Component, Module, and Service

Let's swipe now



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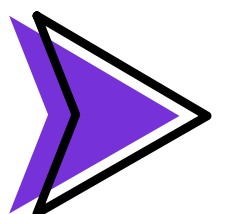
Angular Architecture

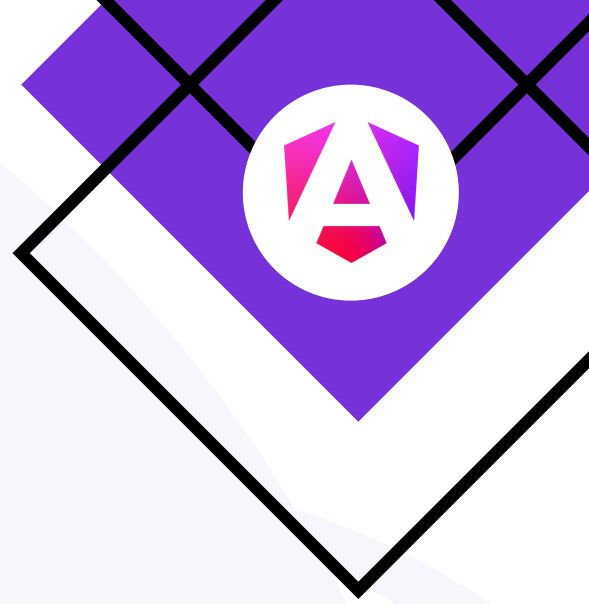
Angular provide a structured way to build and manage parts of an application.

The architecture of Angular is *modular*, meanings that it allows developers to break down applications into smaller and manageable



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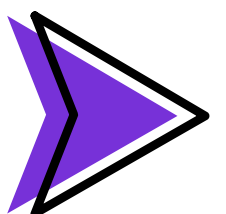


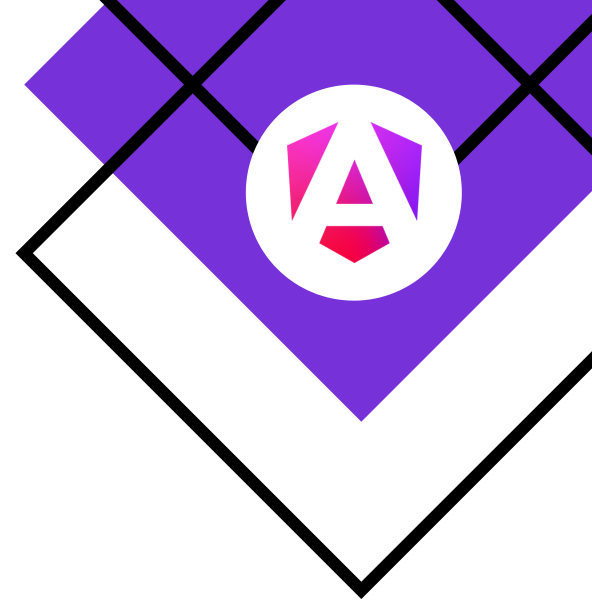
Deep dive

**Exploring the depths of
components, modules, and services**



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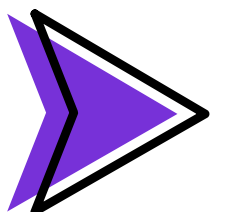


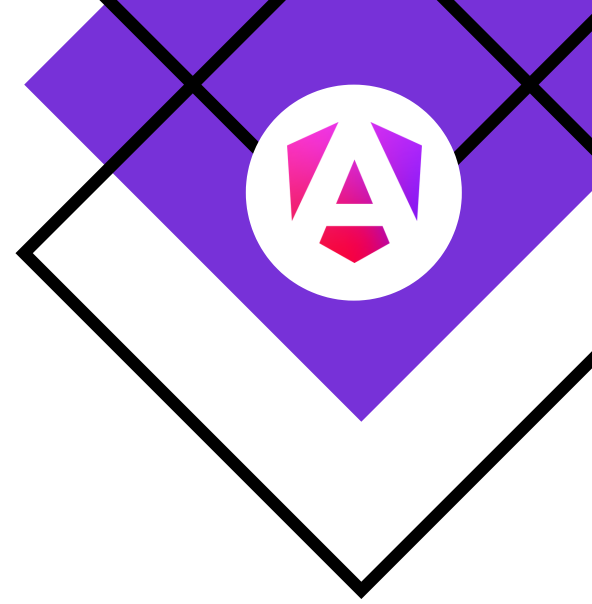


What are Components?



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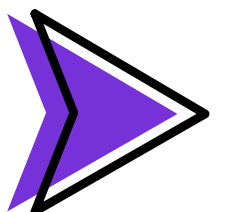


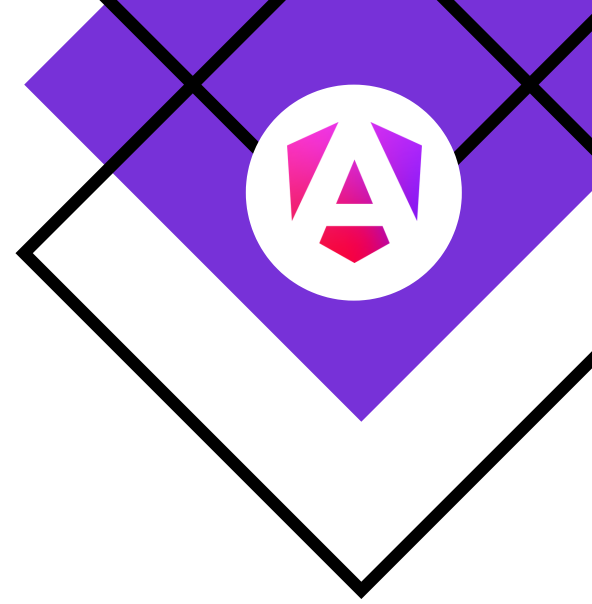
Component

Components are the building blocks of an Angular application. Each component **controls a part** of the user interface (UI) and consists of three main parts.



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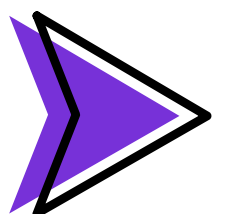


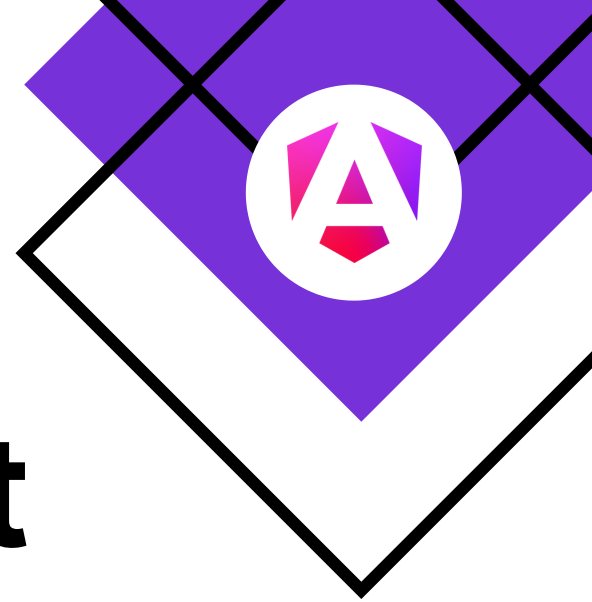
3 Things component must have

- ◆ **Class**, a TypeScript class with behaviors such as handling user input and fetching data from a server for the component.
- ◆ **Template**, define the HTML that controls what renders into the DOM.
- ◆ **Style**, define CSS styles for the component.



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Creating a Component

To create a component, you can use Angular CLI with command `ng generate component` followed by component-name. For example:

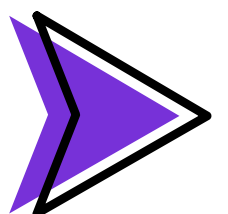
Terminal

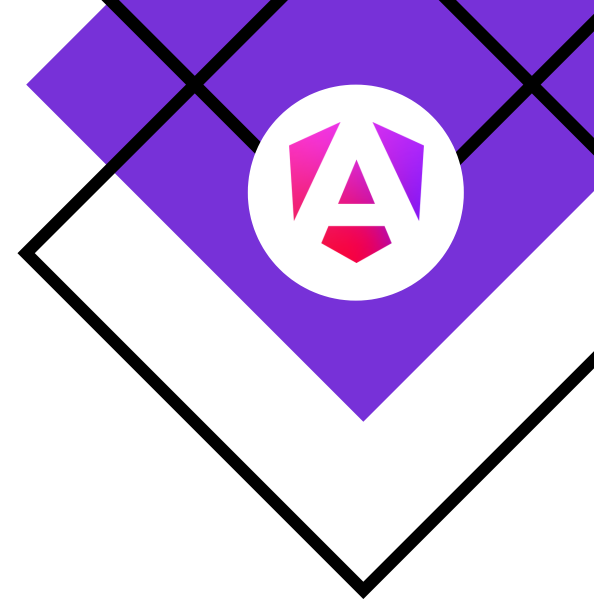
```
ng generate component my-first-component
```

The command will generates a new component with the necessary files.



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Example Component

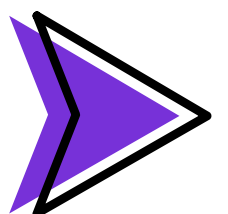
Here's basic component that you have to create.

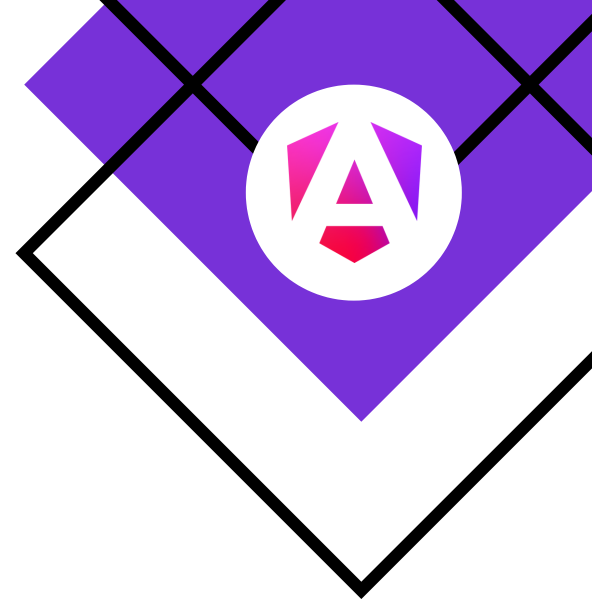
TypeScript

```
1 import { Component } from '@angular/core'
2
3 @Component({
4   selector: 'my-first-component-app',
5   template: '<h1>Hello Angular!</h1>',
6   styles: ['h1 { color: salmon; }']
7 })
8 export class MyFirstComponent {}
```



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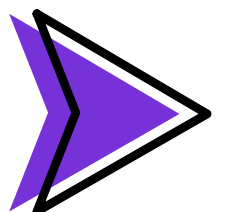


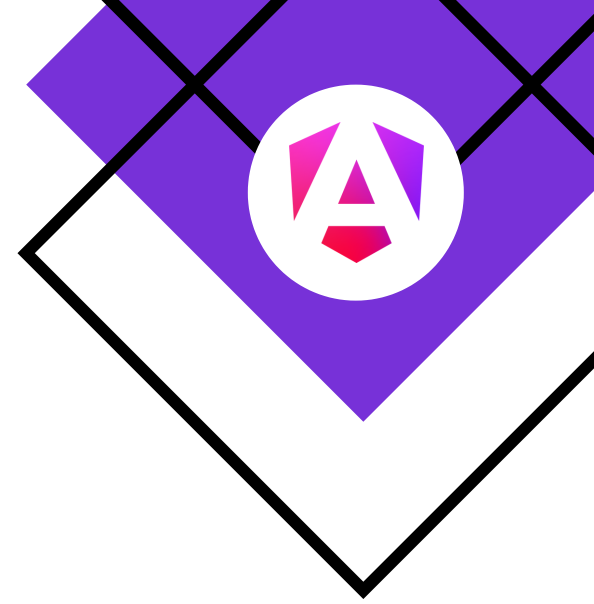


What are Modules?



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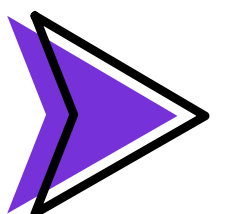


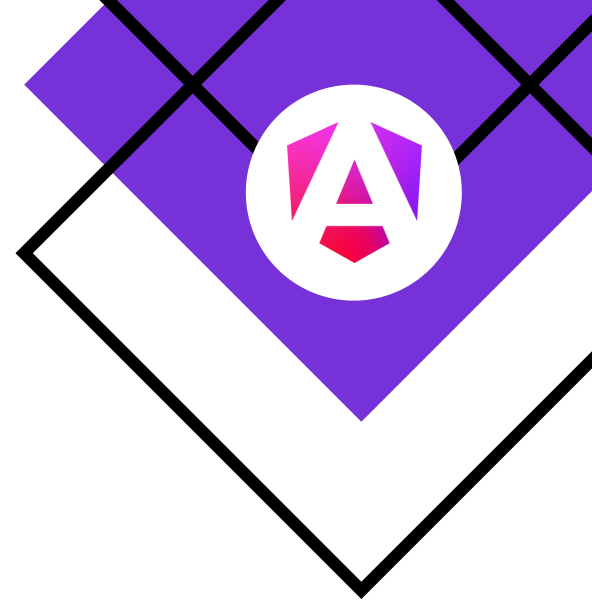
Modules

Modules are containers for a cohesive block of code dedicated to an **application domain**, a **workflow**, or a **closely related set of capabilities**.



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Creating a Module

To create a module, you can use Angular CLI with command `ng generate module` followed by module-name. For example:

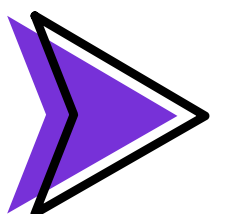
Terminal

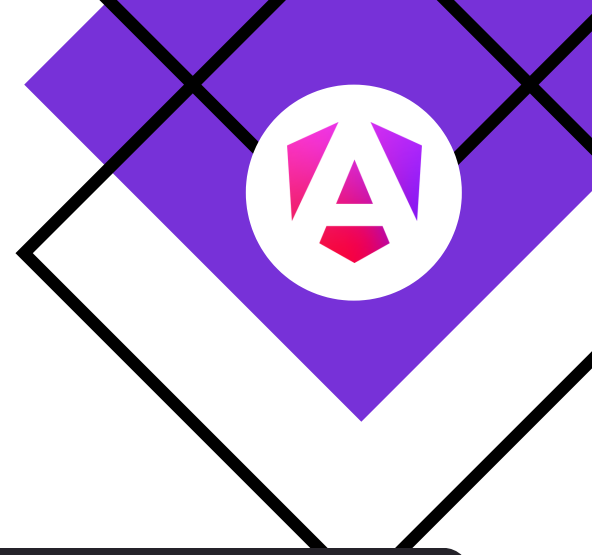
```
ng generate module my-first-module
```

The command will generate a new module. You can import other modules, declare component, and provide services.



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Example Module

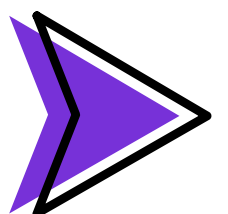
TypeScript

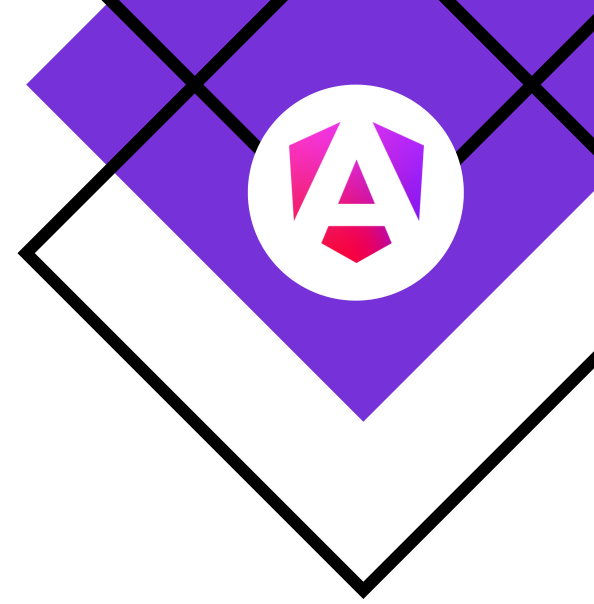
```
1 import { NgModule } from '@angular/core';
2 import { MyFirstComponent } from './my-first-
  component/my-first-component.component';
3 import { MyFirstService } from './my-first-
  service.service';

4 @NgModule({
5   declarations: [MyFirstComponent],
6   providers: [MyFirstService],
7   bootstrap: [MyFirstComponent]
8 })
9 export class AppModule {}
```



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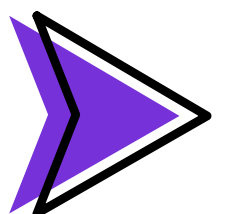
Feature Modules

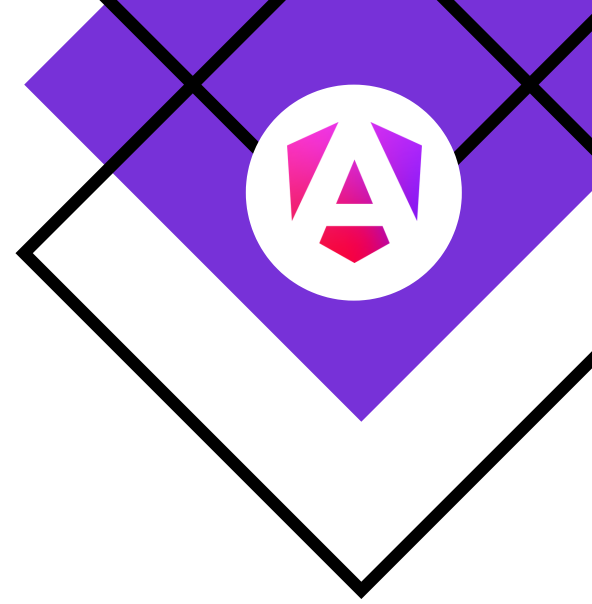
In larger applications, it's best way to create *module* to encapsulate related components, services, and other code.

!NOTE: In Angular v19 recommend using *standalone components* instead of *NgModule* for all new code. Because, by default Angular components are standalone



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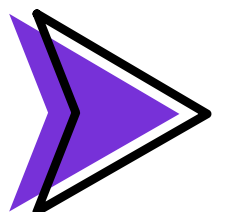


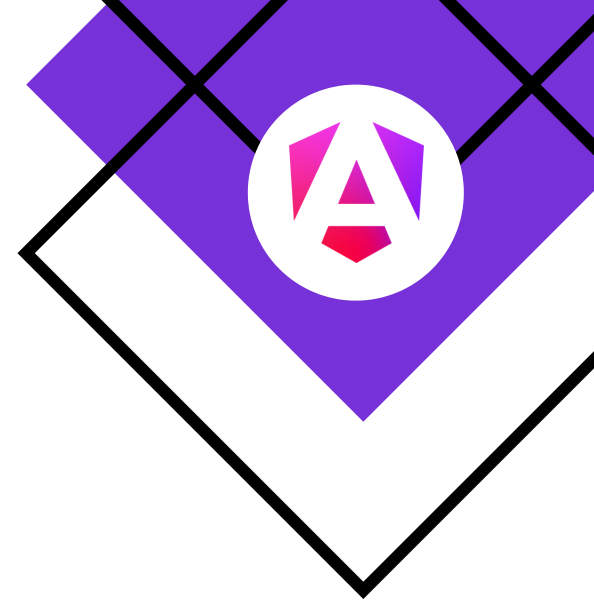


What are Services?



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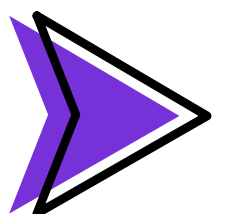


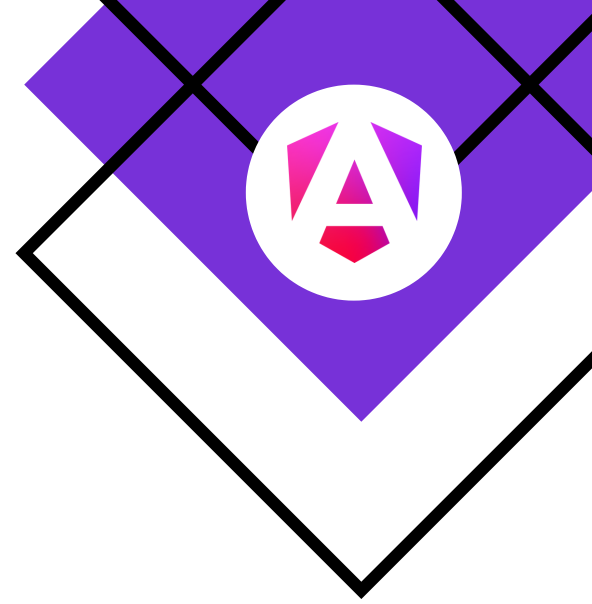
Services

Services are classes that encapsulate **business logic, data access, or any functionality** that can be shared across components. They promote code **reusability** and **separation of concerns**.



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Creating a Service

To create a service, you can use Angular CLI with command `ng generate service` followed by service-name. For example:

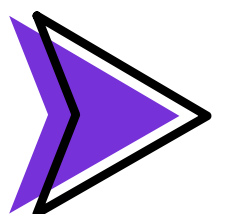
Terminal

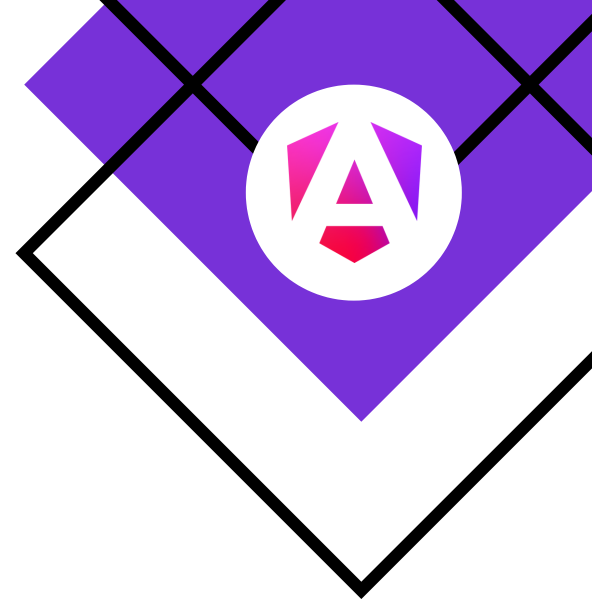
```
ng generate service my-first-service
```

The service is created with the `@Injectable` decorator and a class to write your code.



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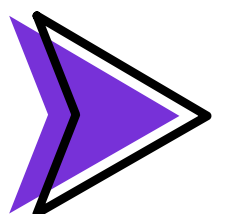
Example Service

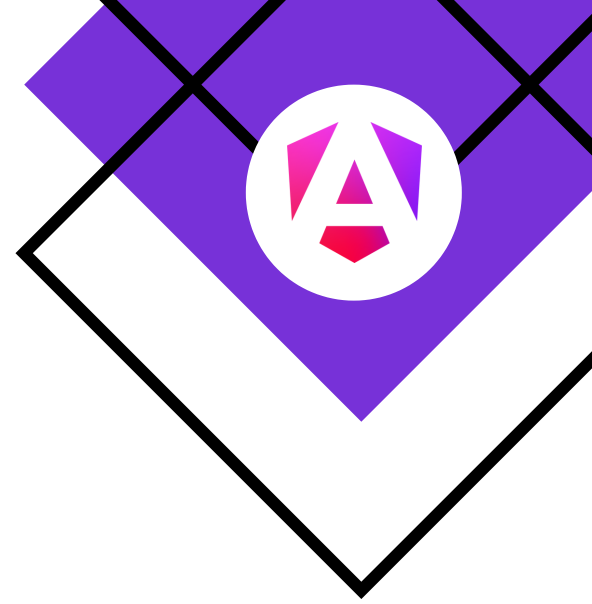
TypeScript

```
1 import { Injectable } from '@angular/core';
2
3 @Injectable({
4   providedIn: 'root',
5 })
6 export class MyFirstService {
7   getFruits() {
8     return ['Apple', 'Orange', 'Watermelon']
9   }
}
```



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Conclusion

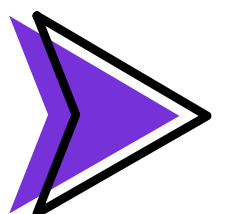
Great job! 🎉

You've just learned the basics of components, modules, and services. These concepts will help you build well-structured applications that are easy to manage and grow.

Keep these ideas in mind, as we'll dive deeper into them soon!



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Don't hesitate to share your thoughts in the comments or DM me anytime!



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